gnani

Intelligence that truly listens.

CAPSTONE BRIEF

VoiceOps 2026

LLM & Generative AI Course · Consulting & Analytics Club, IIT Guwahati × Gnani.ai

Turn what you learned in the last three weeks into a customer support agent people can talk to. You will build a chatbot that answers from company documents and holds a conversation. Then you give it a voice with Gnani's speech models, so customers can simply call and ask.

BEFORE YOU START

Important links

WHAT	LINK
Create your Gnani account and API key	app.gnani.ai/voice
Gnani API quick start	docs.gnani.ai: Quick Start
Gnani Python SDK	docs.gnani.ai: Python SDK
Speech to text reference	docs.gnani.ai: STT REST
Text to speech reference	docs.gnani.ai: TTS REST

Capstone dataset	Capstone_Project_Dataset_FTTT_2026.zip

01 The challenge

You are the AI engineer at ZentroMart, a fictional Indian online store selling groceries and electronics. Every day its support team answers the same questions about late deliveries, returns, refunds and cancellations. Customers want quick, correct answers, and most of them would rather call than type.

Your job is to build ZentroMart's support agent in four steps. Steps 1 to 3 use exactly what the course taught you, from prompting an LLM to RAG over documents. Step 4 adds the ears and the voice, so the same agent can take a phone call.

What ZentroMart gives you

ASSET	CONTENTS
Company documents	Return, refund, delivery, cancellation and warranty policies; product catalog; FAQs; when to hand over to a human
Mock order records	Synthetic customer and order records in CSV tables
Mock tools	`lookup_order(order_id)` and `create_ticket(summary)`
Practice questions	Single questions and multi-turn conversations, with expected decisions and ASR-like text (no audio files)

- You will be tested on new questions your agent has never seen. **Build it to read the documents. Do not hardcode answers.**

Example conversation

1. Customer calls: **"My order still hasn't arrived. Can I cancel it?"**
2. The agent introduces itself as an AI assistant and asks for the order ID.
3. It looks up the order, sees it has already shipped, and explains the cancellation policy for shipped orders.
4. Customer: **"How long will the refund take?"** The agent knows this is about the same order and gives the refund timeline.
5. Customer: **"I want ₹500 compensation right now."** The policy does not allow it. The agent says so politely, creates a ticket, and hands over to a human with a short summary.

Dataset & database guide

Use the fictional company data to ground answers, look up orders, and test the same support agent across text and voice.

DOWNLOAD THE DATASET

Capstone_Project_Dataset_FTTT_2026.zip

Download, extract, then begin with README.md and DATA_DICTIONARY.md.

IN THE ZIP

Files and their purpose

FOLDER / FILES	HOW TO USE IT
README.md	Start with the project setup and read the field definitions in DATA_DICTIONARY.md.
kb/ · policy docs	Chunk and index Markdown documents for RAG. Prioritize pol_returns_v4.md; pol_returns_v3_superseded.md is an older version.
data/ · CSV tables	Customers, orders, order items, products, support logs, pincodes and holidays. Use the dictionary to identify keys and join related records for lookups.
tools/ · Python	Inspect the provided helper modules; connect order lookup and ticket creation to the conversational agent.
practice/ · CSV	Use practice questions and conversations to measure prompting-only answers, then RAG and multi-turn quality.

RECOMMENDED WORKFLOW

1. **Load the knowledge base**
   Index the current policies and FAQs; save retrieved source chunks.
2. **Build the data and tool layer**
   Read the CSV schema and implement real lookups / ticket actions.
3. **Evaluate and add voice**
   Compare Step 1 vs Step 2; test follow-ups, then add Gnani STT/TTS.

Dataset schema

Nine CSV files, 21 knowledge-base documents and two Python mock tools. Use the verified table counts below and keep all IDs as strings.

RELATIONAL STRUCTURE

Customers customer_id → **Orders** customer_id, order_id → **Order items** order_id, sku → **Products** sku

Reference joins: `orders.delivery_pincode → pincodes.pincode`. Use `holidays.date` for business-day timelines.

DATA / SEVEN CSVS

Core database tables

TABLE / ROWS	KEYS & JOINS	USEFUL FIELDS
customers.csv 300 customers	customer_id	phone, pincode, is_plus, preferred_language
orders.csv 1,000 orders	order_id FK: customer_id	status, promised_date, payment, return and refund fields
order_items.csv 1,783 line items	order_id + line_no FK: sku	qty, unit_price_inr, return_category
products.csv 70 products	sku	price_inr, fulfilled_by, assured, warranty_months
pincodes.csv 36 pincodes	pincode	serviceable, cod_available, express_available
holidays.csv 4 dates	date	name; skip holidays when counting business days
support_logs.csv 244 historical logs	log_id customer_id, order_id	transcript, intent, resolution; context only

kb/ · 21 Markdown documents
Policies, agent guides, help and FAQs. Index by section ID for RAG.

practice/ · 2 CSV files
52 questions + 23 conversations (96 customer turns) for evaluation.

How to use the dataset

Follow this workflow to build a grounded support agent for ZentroMart, from data ingestion to voice testing.

DOWNLOAD THE UPDATED DATASET

Capstone_Project_Dataset_FTTT_2026.zip ↗

Extract the ZIP. Read **README.md** and **DATA_DICTIONARY.md** first.

RECOMMENDED WORKFLOW

1 Load the CSVs and map relationships

Use `pd.read_csv(path, dtype=str)`. Join orders to customers using `customer_id`; join items to orders with `order_id`, and items to products with `sku`. Empty fields mean missing or not applicable.

2 Build your knowledge-base retrieval

Chunk `kb/*.md` and preserve each `[SECTION-ID]`, `status` and `effective_from`. Follow precedence: current policy → agent guide → help → FAQ → marketing. Historical support logs are **not** policy authority.

3 Connect tools and verify the caller

Import `lookup_order` and `create_ticket` from `tools`. Before revealing order details, match the customer's phone's last 4 digits **or** delivery pincode. Confirm a handover only after `created=True`.

4 Evaluate on practice cases

Use `practice/questions.csv` to compare prompt-only vs. RAG answers; `asr_text` simulates transcription. Process `practice/conversations.csv` by `conversation_id` and `turn`. Check expected decisions, tool calls and `gold_evidence`.

5 Apply date rules, then add speech

Treat **20 October 2026** as today's date. Business days run Mon–Sat, excluding `holidays.csv`; return windows use calendar days. No audio clips are included: test with your own recorded voice and Gnani STT/TTS.

Important: This refined ZIP uses **ZentroMart** consistently. The `lookup_order` tool enforces caller verification; run `python validate_dataset.py` before building.

ZIP structure & file schema

Use this as a map of the supplied archive: which folders contain knowledge, structured data, practice cases and executable tools.

Folder structure

```
Capstone_Project_Dataset_FTTT_2026/
├── README.md
├── DATA_DICTIONARY.md
├── ZIP_SCHEMA.md
├── CHANGELOG.md
├── validate_dataset.py
├── kb/                      21 .md files
│   ├── 00_about_this_knowledge_base.md
│   ├── agent_*.md
│   ├── pol_*.md
│   ├── help_*.md
│   ├── faq_*.md
│   └── mkt_*.md
├── data/                    7 .csv files
│   ├── customers.csv          300
│   ├── orders.csv           1,000
│   ├── order_items.csv      1,783
│   ├── products.csv            70
│   ├── pincodes.csv            36
│   ├── holidays.csv             4
│   └── support_logs.csv       244
├── practice/                2 .csv files
│   ├── questions.csv           52
│   └── conversations.csv       96 turns
└── tools/
    ├── __init__.py
    └── tools.py
```

The practice set includes 23 conversations (96 turns). Use `ZIP_SCHEMA.md` to check relationships and run `validate_dataset.py` to validate the extracted files.

Schema & primary keys

Customers	`customer_id`
Orders	`order_id`
Order items	`order_id` + `line_no`
Products	`sku`
Pincodes	`pincode`
Holidays	`date`
Support logs	`log_id`

Treat IDs and pincodes as strings. Blank fields mean missing / not applicable. Read `DATA_DICTIONARY.md` for every column and `ZIP_SCHEMA.md` for the joins.

HOW THE TABLES CONNECT

customers.customer_id = **orders.customer_id** · **orders.order_id** = **order_items.order_id**

order_items.sku = **products.sku** · **orders.delivery_pincode** = **pincodes.pincode**

Use **holidays.date** as a calendar reference for business-day calculations.

Usage: Index `kb/*.md` for RAG; read `data/*.csv` for factual lookups; run `tools/` for order lookup and ticket creation; evaluate with `practice/*.csv`. Use **20 Oct 2026** as the dataset's reference date. The practice files contain transcripts, not audio recordings. Start with `README.md`.

VOICEOPS AGENT PIPELINE · ONE TURN



02 The four steps

Each step adds one thing to the same agent, and each one reuses a week of the course. Finish all four to submit.

1 Answer a question with an LLM

RECAPS WEEKS 1 AND 2: YOUR FIRST LLM API CALL AND PROMPTING

Call any LLM through its API, the way you did in the Week 1 checkpoint and the Week 2 build. Give it a system prompt that makes it ZentroMart's support assistant, then send it a customer question.

- The system prompt sets the assistant's role and limits.
- The assistant's first reply tells the customer they are talking to an AI assistant.
- Try it on the practice questions ZentroMart gives you.

At this point it has not seen ZentroMart's documents, so it will guess. Note how many practice questions it gets right. **Step 2 should beat that number.**

2 Ground it in the documents

RECAPS WEEK 3: RAG

Give the assistant ZentroMart's documents so it answers from facts.

- Split the documents into chunks and store their embeddings in a vector index.
- For each question, retrieve the most relevant chunks and answer only from them.
- When the documents do not cover a question, say so. Never invent a policy, date or amount.
- Keep track of which chunks each answer used.

Run the same practice questions again and compare with Step 1. **Put both numbers in your report.**

3 Hold a conversation

RECAPS WEEKS 2 AND 3: CONTEXT WINDOWS AND EVALUATION

Turn single answers into a real support conversation.

- Remember what the customer already said, such as the order ID and the problem.
- Understand follow-ups such as "how long will that take?" without asking the customer to repeat.
- Ask one clear question when a needed detail is missing.
- Use `lookup_order` and `create_ticket` when needed. Never say you did something your code did not do.
- Politely refuse requests that break policy, and hand over to a human with a short summary when the documents say a person must decide.

Try it on the practice conversations ZentroMart gives you.

4 Give it a voice

THE NATURAL NEXT STEP: REAL CUSTOMERS CALL SUPPORT

Wrap your Step 3 agent with Gnani's speech models. The agent's brain stays the same. Customer audio goes through Gnani STT into your agent, and the agent's reply comes back out through Gnani TTS.

To set up

1. Create a Gnani account and generate an API key at app.gnani.ai/voice. One key covers both STT and TTS.
2. Save the key as the environment variable `GNANI_API_KEY`. Never put it in your code or commit it to Git.
3. Install Gnani's official Python SDK: `pip install gnani-vachana`.

Then the voice layer is about ten lines:

```
from gnani.stt import GnaniSTTClient
from gnani.tts import GnaniTTSClient

stt = GnaniSTTClient()  # reads GNANI_API_KEY from your environment
tts = GnaniTTSClient()

def voice_turn(audio_path, agent):
    heard = stt.transcribe(audio_path, language_code="en-IN")["transcript"]
    if not heard.strip():
        reply = "Sorry, I didn't catch that. Could you say it again?"
    else:
        reply = agent.respond(heard)  # your Step 3 agent
    tts.synthesize(reply, voice="Kaveri", model="timbre-v2.5",
                   language="en-IN", output_file="reply.wav")
    return heard, reply
```

STEP 4 · GIVE IT A VOICE

Things to know

The two calls come from the Python SDK page. Things to know, each from its page on docs.gnani.ai:

Always name the TTS model. Python SDK · Voice catalog

The SDK's older default is no longer served and returns an error. The sample uses `timbre-v2.5`. `timbre-v3.0` is also available, with 13 voices that each speak all 10 languages.

Keep each customer turn short. STT REST

The REST STT endpoint accepts clips up to 60 seconds.

Use any of 10 Indian languages. STT REST · TTS REST

Gnani STT handles Bengali, English, Gujarati, Hindi, Kannada, Malayalam, Marathi, Punjabi, Tamil and Telugu; set `language_code` on each call, for example `gu-IN`. TTS speaks the same languages plus `hi-en` for Hinglish. Your agent can listen in one language and reply in another, such as hearing Gujarati and answering in Hindi.

Mixed English and Hindi is fine. STT Realtime

The `en-IN` option accepts English mixed with Hindi and returns Latin script.

Write replies for the ear. Text normalization guide

Include the year in dates, and break long order numbers into groups with spaces so they are read clearly.

Pick a voice. Voice catalog

English voices for `timbre-v2.5` include Kaveri, Trupti, Pranav and Shlok.

- When it works, record a 2-minute demo of a spoken conversation.

03 Rules

Use Gnani for listening and speaking. Everything else is your choice.

COMPONENT	RULE
Speech to text (STT)	**Gnani API only**
Text to speech (TTS)	**Gnani API only**
LLM	Any hosted or open-source LLM
Embeddings, vector database, search	Any
Framework	Any, or plain Python
Interface	Anything that plays and records audio: a web page or a notebook

- No other speech-to-text or text-to-speech service is allowed, even as a backup.
- Use only the synthetic customer data provided. Never use real customer data or real call recordings anywhere in building your solution.
- You may record your own voice for testing. Do not record other people without their written consent.
- Each participant uses their own Gnani account.
- Your agent must answer from the documents. Hardcoded answers will fail the test questions.
- Test answers must come from your agent, never typed or edited by hand.

04 How you're evaluated

Most of your score comes from what the course taught: grounded answers and a well-handled conversation. Voice is worth 20 marks.

HOW IT WORKS

1

Practice on ZentroMart's practice questions and conversations as often as you like.

2

Submit your code with a run command (see What to submit).

3

The organizers run every agent on the same new test questions, recorded as audio, and score the answers automatically.

4

The top 50 demo their agent live to the judges.

Rubric (100 marks)

The first five rows are scored automatically for every participant. The last two are judged for the top 50. **Using any speech service other than Gnani's disqualifies a submission.**

CRITERION	MARKS	WHAT WE CHECK	STEP
Correct, grounded answers	25	Answers match the documents, and the agent says "I don't know" when they don't cover the question	2
Retrieval quality	15	The agent found the right document chunks	2
Conversation handling	20	Follow-ups, clarifying questions, refusals and handovers to a human	3
Your own evaluation	10	Your Step 1 vs Step 2 comparison and what you learned from it	1 and 2
Voice works end to end	10	The agent handles every test audio file and replies with speech	4
Voice experience	10	Judges listen: clear, short replies and recovery from unclear audio	4
Report and demo	10	A short report and a demo that shows the agent working	All
Total	100	Sum of all seven criteria above	

05 What to submit

Code repository	Public Git repo with a README explaining how to install and run your agent, plus `.env.example` with no keys in it
Run command	One command that takes a folder of customer audio files, runs your agent on each, and saves its replies (as text and audio) with the document chunks it used. The organizers use this to test every agent the same way.
Short report	Max 4 pages: what you built in each step, your Step 1 vs Step 2 scores, what went wrong and what you would fix, any bonus work
Demo video	Max 2 minutes: a question answered from the documents with a follow-up, plus one handover or "I don't know"

Submission ZIP structure

Package these deliverables together so your agent can be tested reproducibly.

team_name_voiceops.zip

```
|-- README.md
|-- .env.example
|-- requirements.txt *
|-- run.py / run.sh *
|-- src/
|-- report.pdf
|-- demo.mp4
`-- sample_outputs/
```

* Equivalent files accepted for other stacks.

HOW TO SUBMIT

README + repository
Include the public GitHub URL, setup steps and exact run command.

Batch run
Accept a folder of audio; save text, speech output and retrieved sources.

Keep secrets out
Do not include .env, API keys, virtual environments or a dataset copy.

All the best,
Team C&A × Gnani.ai

gnani